Vinyl-Chloride Process Safety and Environmental Hazard Analysis

CHE 396 Senior Design Mini Project 4

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Table of Contents

1. Project Charter

Project Objectives

- Identify major safety hazards associated wih vinyl-chloride production process.
- Evaluate environmental concerns, including air emissions, wastewater impacts, and solid waste production.
- Assess risks associated with operating conditions to determine potential failure points and develop mitigation strategies.
- Conduct a HAZOP study to propose more safety recommendations

Scope of Work

- Detailed review and summary of process flow diagram.
- Hazard identification (HAZOP) of vinyl-chloride production process with special emphasis on T-100 process units.
- Analysis of temperature and pressure hazards based on operation conditions.
- Identification of OSHA PSM highly hazardous materials.
- Evaluation of environmental risks.
- Development of preventative measures and mitigation systems for identified hazards.

Deliverables

- Project charter and design plan
- Environmental and Safety Assessment for T-100 Process Units
- Hazop Study
- Mitigation recomendations
- AI Generated analysis
- Comprehensive final report

1.1 Roles and Responsibilities

Project Coordinator	Defines project objectives, scpe, and deliverables.
	Develops the Project Charter and Design Plan.
	Perfoms final editing.
Planner and AI Integration Evaluator	Creates Gantt chart.
	Conducts AI process safety analysis.
	Performs and AI performance evaluation.
Environmental Analyst	Identifies environmental concerns such as emissions, wastewater, and waste generation.
	Proposes design features and engineering controls for mitigating environmental impacts.
Safety Analyst	Identifies OSHA PSM chemicals.
	Evaluates toxic, flammable, reactive, and thermal hazards.
	Recommends safeguards such as relief valves and shutdown systems.
Process Risk Analyst (HAZOP)	Conducts a HAZOP study on T-100 units.
	Identifies causes and consequences.
	Reviews existing safeguards and proposes additional safety measures.

1.2 Project Timeline

Key Milestone	Description	Deadline
Project Start off	Identify objectives and team roles.	11/27/2025
Process Review	Analyze PFD and stream table.	11/30/2025
Gantt Chart Completion	Set project timeline.	11/30/2025
Environmental Analysis	Identify emissions and waste concerns.	12/02/2025
Process Hazards and Safety Evaluation	Review OSHA chemicals and safety risks.	12/02/2025
HAZOP Study	Assess risks associated with T-100 units.	12/02/2025
AI Safety Review	Analyze AI generated response on hazard identification.	12/02/2025
Draft Report Completion	Combine sections into a full draft.	12/02/2025
Final Editing and Revisions	Finalize formatting and consistency.	12/05/2025
Report Submission	Submit completed project.	12/05/2025

2. Design Plan

2.1 Summary of Process

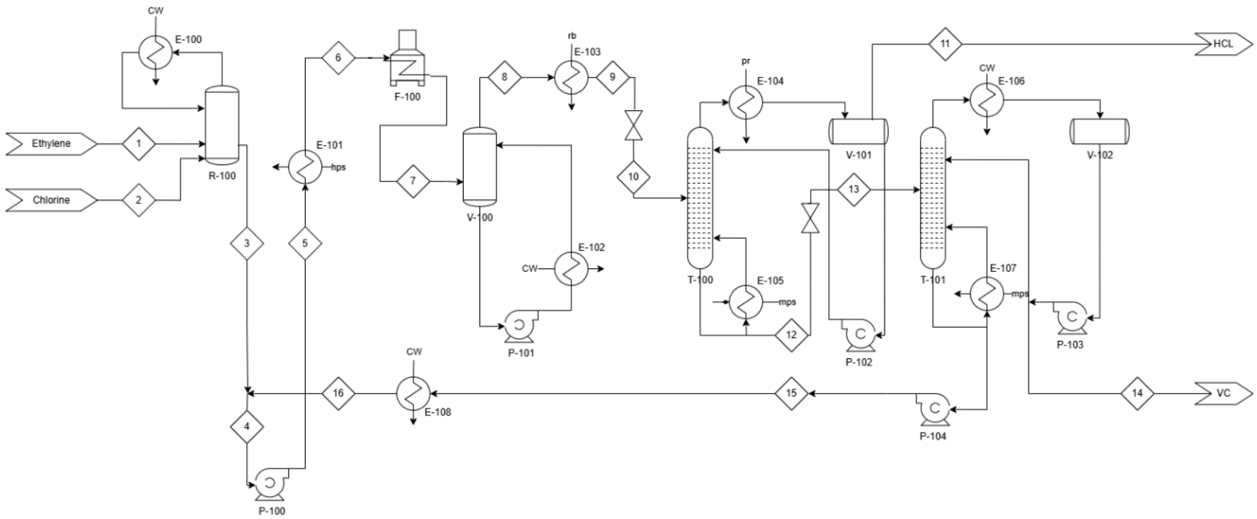


Figure 1: *Process Flow Diagram for Vinyl-chloride Production*

The vinyl chloride production process begins by reacting ethylene with chlorine in the chlorination reactor R-100, where a highly exothermic reaction forms 1,2 dichloroethane (EDC). To maintain safe reaction temperatures, heat is continuously removed through the E-100 cooler, which uses cooling water. The liquid EDC then flows to the E-101 evaporator, where it is vaporized to prepare it for the cracking stage.

The vaporized EDC enters the F-100 pyrolysis furnace, where it undergoes high-temperature thermal cracking to produce vinyl chloride (VC) and hydrogen chloride (HCl). Because this reaction occurs at temperatures near 500°C, precise control is required to avoid unwanted product formation. The hot reaction mixture then flows into the V-100 quench tank, where it is rapidly cooled to stop further cracking reactions. Additional cooling occurs in E-102 and E-103, which condense part of the gaseous mixture, bringing the stream to conditions suitable for separation.

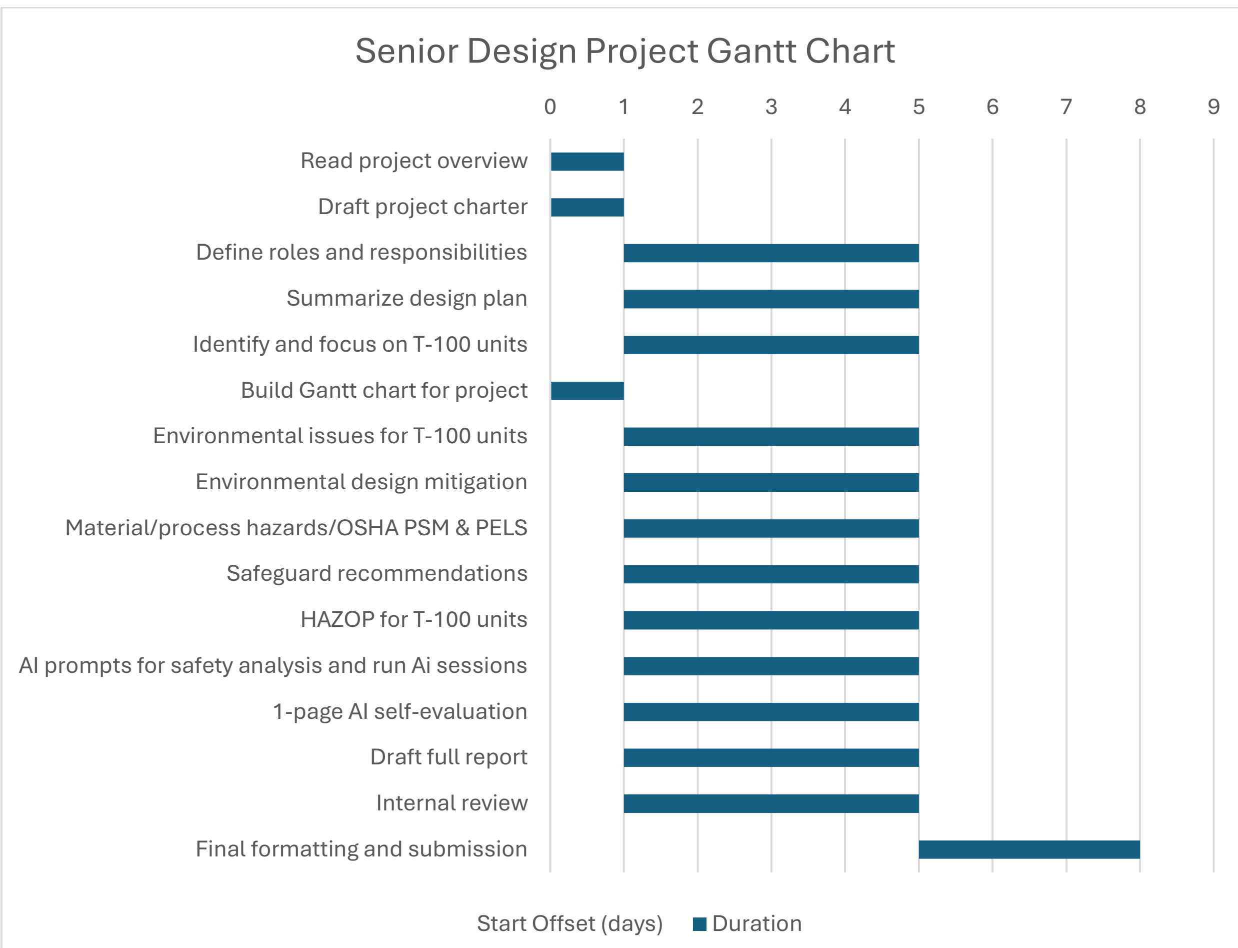
The partially condensed stream enters the T-100 distillation column, where it undergoes the first separation. In T-100, HCl is separated overhead as a light component, while VC and unreacted EDC remain in the heavier liquid phase. The bottom liquid from T-100 is then routed to the second column, T-101, where VC is purified and removed as the overhead product. The remaining by-products are separated at the bottom. The purified VC is condensed in E-106 and collected as the final product, while the unreacted EDC is recycled back to R-100 for reuse, thereby improving process efficiency and reducing waste.

Throughout the process, recycling pumps (P-100, P-101, P-102, P-103, and P-104) maintain a stable flow through the system to ensure continuous operation. Heat exchangers, condensers, and reboilers help maintain the temperature and pressure conditions required for effective separation.

2.2 Focus Areas

1. **T-100 and T-101 Distillation Column:**
    a. Flammability and carcinogenic properties of vinyl chloride.
    b. Corrosion and toxicity hazards from hydrogen chloride.
    c. Over-pressurization, flooding, or tray malfunction.
    d. Leak risks from pumps and valves.
    e. Possible air emissions of vinyl chloride and HCl.
    f. Hazardous components and by-product disposal.
    g. Condenser and reboiler failures.

2.3 Gantt Chart



#	Task	Person	Start Date	End Date
1	Read project overview	Whole Team	11/27/25	12/2/25
2	Draft project charter	Angelica	11/27/25	11/28/25
3	Define roles and responsibilities	Angelica	11/28/25	12/2/25
4	Summarize design plan	Angelica	11/28/25	12/2/25
5	Identify and focus on T-100 units	Angelica	11/28/25	12/2/25
6	Build Gantt chart for project	Ify	11/28/25	12/2/25
7	Environmental issues for T-100 units	Javier	11/28/25	12/2/25
8	Environmental design mitigation	Javier	11/28/25	12/2/25
9	Material/process hazards/OSHA PSM & PELS	Cesearo	11/28/25	12/2/25
10	Safeguard recommendations	Cesearo	11/28/25	12/2/25
11	HAZOP for T-100 units	Hiba	11/28/25	12/2/25
12	AI prompts for safety analysis and run AI sessions	Ify	11/28/25	12/2/25
13	1-page AI self-evaluation	Ify	11/28/25	12/2/25
14	Draft full report	Whole Team	11/28/25	12/2/25
15	Internal review	Whole Team	11/28/25	12/2/25
16	Final formatting and submission	Hiba	12/2/25	12/5/25

3. Environmental And Safety Assessment For T-100 Process Units

3.1 Environmental Issues

Environmental hazards can be present in the production of PVC processing, resulting from air emission pollution, wastewater contamination, and solid waste contamination.

3.1.1 Environmental Concerns

The concerns of air pollution in the environment fall under damaging the ecosystem with toxic and flammable gases. Exposure to air pollution from PVC processing can also lead to respiratory, carcinogenic, and long-term health problems.

Table 1: *Air Pollutants*

Pollutant	Harm to the environment and the surroundings
Vinyl Chloride	Carcinogenic, Asphyxiation, Toxic Gas.
1,2-Dichloroethane	Carcinogenic, Asphyxiation, Flammable Gas.
Ethylene Gas	Flammable Gas.
Chlorine Gas	Reactive, Toxic Gas.
Hydrogen Chloric Acid Vapor	Corrosive Gas.

Possible sources of pollutants from air emissions can come from leaks and improper separation from the following sources.

Table 2: *Air Pollutant Sources*

Source	Pollutant present
Stream 1	Ethylene.
Stream 2	Chlorine.
Stream 6	1,2-Dichloroethane.
Stream 7	1,2-Dichloroethane, Vinyl Chloride, HCl.
Stream 8	Vinyl Chloride, Hydrogen Chloride
Stream 11	HCl
Stream 14	Vinyl Chloride.
Feed Tanks (V-100)	Chlorine, Ethylene.
Furnace (F-100)	1,2-Dichloroethane, Vinyl chloride, HCl.
Quench Cooler (E-100)	Vinyl chloride, HCl.
Distillation Column (T-100)	HCl, Vinyl chloride.
Condenser (E-101)	Chlorine, Ethylene. 1,2-Dichloroethane, Vinyl Chloride, HCl.
Condenser (E-102)	1,2-Dichloroethane, Vinyl Chloride, HCl.
Furnace Vent (F-100)	CO2, Chlorine, Ethylene.

The concerns of Wastewater contamination pollution in the environment fall under damaging the ecosystem with toxic and corrosive liquids. Exposure to Wastewater pollution from PVC processing can also lead to Carcinogenic and long-term health problems.

Table 3: *Wastewater Pollutants*

Pollutant	Harm to the environment
Vinyl Chloride	Carcinogenic, Toxic liquid.
1,2-Dichloroethane	Carcinogenic, Flammable liquid.
Hydrogen Chloric Acid Vapor	Corrosive liquid.
Cooling Water	Contains traces of Chlorine.

Possible sources of Pollutants from wastewater can come from leaks and improper separation form the following sources.

Table 4: Wastewater Pollutant Sources	
Source	**Pollutant present**
Stream 3	1,2-Dichloroethane
Stream 4	1,2-Dichloroethane
Stream 5	1,2-Dichloroethane.
Stream 9	1,2-Dichloroethane, Vinyl chloride, HCl
Stream 10	1,2-Dichloroethane, Vinyl chloride, HCl
Stream 12	1,2-Dichloroethane, Vinyl chloride
Stream 13	1,2-Dichloroethane, Vinyl chloride
Stream 15	1,2-Dichloroethane
Stream 16	1,2-Dichloroethane
Reactor (R-100)	1,2-Dichloroethane
Pumps (P-100)	HCl, Vinyl chloride.
Pumps (P-101)	1,2-Dichloroethane, Vinyl Chloride, HCl.
Pumps (P-102)	1,2-Dichloroethane, Vinyl Chloride.
Distillation Column (T-100)	1,2-DCE, Vinyl chloride, HCl
Condensers (E-101)	HCl.
Condensers (E-102)	Vinyl chloride
Reboiler	Chlorine, Ethylene. 1,2-Dichloroethane, Vinyl Chloride, HCl.
Heat Exchangers	Cooling Water
Wastewater Systems	Traces of Chlorine, Ethylene. 1,2-Dichloroethane, Vinyl Chloride, HCl.

The concerns of solid contamination pollution in the environment include damaging the ecosystem with toxic and cancer-causing solids. Exposure to solid pollution from PVC processing can also lead to Carcinogenic and long-term health problems.

Table 5: Solid Pollutants	
Pollutant	**Harm to the environment**
Vinyl Chloride	Toxic solids in long-term exposure.
1,2-Dichloroethane	Carcinogenic Solid in long-term exposure.
Residue of HCl and Vinyl chloride	Carcinogenic and Toxic solids in long-term exposure.
Burning residue	Carcinogenic Solid.

Table 6: *Solid Pollutant Sources*	
Source:	**Pollutant present:**
Stream 11	HCl.
Stream 14	1,2-Dichloroethane, Vinyl Chloride.
Distillation Column (T-100)	Residue of HCl and Vinyl chloride.
Furnace Gas Vent (F-100)	Burning residue.

3.1.2 Engineering Solutions

Table 7: *Air Pollution Engineering Solutions*		
Pollutant Source	**Method of Control**	**Engineering solutions**
HCl in T-100 and stream 11,	Scrubbing	Neutralize HCl with water and a neutralizing agent
Vinyl Chloride from T-100 and in Stream 14.	Liquify and absorb in a Packed bed of Activated charcoal.	Liquify the Vinyl Chloride and treat it with activated charcoal to create a disposable product
Leak Detection syst	Leak prevention and monitoring	Apply a Leak Detection and Repair program to identify leaks for immediate repair
All Streams (Chlorine, Ethylene. 1,2-Dichloroethane, Vinyl Chloride, HCl. And CO2)	Regulation of pressure	The system regulates pressure and controls the flow of gas to prevent damage to the streams.
Furnace (F-100) Vent shaft	Thermal oxidation	Use a Thermal Oxidizer to remove all soot and residue from burning.

Table 8: *Wastewater Contamination Engineering Solutions*		
Pollutant Source	**Method of Control**	**Engineering solutions**
Acidic Wastewater	PH Neutralization	Neutralize PH of Acidic wastewater with neutralizing solutions like NaOH. To dispose of the solution
Solids in Wastewater	Filtration	Filter out solid monomers from the wastewater before disposing,
Spill containment	Preventive containment,	Prepare an environment that can contain spills of chemicals through tanks and drains to prevent spilling into the environment.

Cooling water	Closed-loop cooling	Recycle Cooling Water to limit the use of fresh water.

Table 9: *Solid Contamination Pollution Engineering Solutions*		
Pollutant Source	**Method of Control**	**Engineering solutions**
Furnace vent shaft	Scrubbing	Use scrubbing tools regularly to prevent clogging of the vents from solid residue from burning.
Unwanted Solids	Disposal	Filter out solids from wastewater and dispose of solids through a disposal company

Table 10: *General Engineering Solutions*		
Method of Control	**Equipment and Tools**	**Application**
Continuous emission monitoring (CEMS)	HCl, VCM, Cl_2 analyzers on gas and liquid emissions	Used to create a manifest for awareness in maintain pollutants within EPA amounts.
PH sensitivity	PH sensors in streams	Monitors pH changes to detect unsafe pH levels.
Leak detection software	Ultrasonic detectors, IR cameras, and Pressure sensors.	Early warning for fugitive emissions.
Automated shutdown	Pressure regulated	Limits Pressure to a safe amount to prevent over-pressurization.
Containment	Tanks, Drainage systems, Containment chambers.	Used to limit environmental exposure during leaks.

3.2 Process Hazards and Safety Compliance

3.2.1 Material and Process Hazards

Within the T-100/T-101 process units, the primary components present in the inlet and outlet streams are 1,2-dichloroethane, vinyl chloride, and hydrogen chloride. Together, these substances form a highly volatile and corrosive mixture that poses significant operational hazards. Of these, hydrogen chloride is specifically listed under OSHA's Process Safety Management (PSM) standard as a highly hazardous chemical. Its permissible exposure limit (PEL) ranges from approximately 0.3 ppm to 5.0 ppm, depending on regulatory jurisdiction, industry sector, and workplace conditions, underscoring the need for stringent monitoring and control measures.

Section 8. Exposure controls/personal protection

Control parameters

Occupational exposure limits

Ingredient name	Exposure limits
hydrogen chloride	**California PEL for Chemical Contaminants (*Table AC-1*) (United States).** PEL: 0.3 ppm 8 hours. CEIL: 2 ppm **ACGIH TLV (United States, 3/2020).** C: 2 ppm **OSHA PEL 1989 (United States, 3/1989).** CEIL: 5 ppm CEIL: 7 mg/m³ **NIOSH REL (United States, 10/2016).** CEIL: 5 ppm CEIL: 7 mg/m³ **OSHA PEL (United States, 5/2018).** CEIL: 5 ppm CEIL: 7 mg/m³

Figure 2: *Exposure Control/Personal Protection for HCl, obtained from "Safety Data Sheet: HCl". Accessed December 1, 2025. https://www.airgas.com/msds/001028.pdf.*

Hazard #1

For the overall process, high temperatures aren't an issue; however, stream 11, leaving T-100, gets much colder than any other stream in the process, which could potentially cause embrittlement and/or stress cracking in the metal piping that carries the cold, gaseous, eventually condensed HCl.

Hazard #2

Since T-100 involves the separation of HCl, which is corrosive, the trays and the column can possibly experience deterioration and/or other physical damage. Unit T-100 also contains two toxic and highly flammable fluids, VC and 1,2-dichloroethane. Without the proper temperature and pressure, these fluids can combust, leading to an explosion. As these two fluids make their way to T-101, they present the same process hazards.

Hazard #3

Both distillation columns present a potential risk of fluid leakage into the plant or the surrounding environment. Such releases may result from inadequate maintenance practices, deterioration of seals and gaskets at flanged connections, or internal mechanical failures within the column. These scenarios pose significant safety and health hazards, necessitating proactive inspections, preventive maintenance, and robust containment measures to mitigate risk.

3.2.2 Risk Assessment

Risk #1

Given the high flammability of several process components, minimizing potential ignition sources throughout the facility is essential. Careful consideration of each material's autoignition temperature and flashpoint must guide operating practices to reduce the risk of fire or explosion. Continuous temperature monitoring, achieved through strategically placed thermocouples, coupled with automated alarm systems that trigger when critical thresholds are reached, will provide early warnings and enable rapid intervention.

Risk #2

Because of the strong corrosivity of hydrogen chloride (HCl), the T-100 unit and associated piping for stream 11 should be constructed from materials specifically resistant to acidic environments. In addition, since stream 11 involves sub-zero gaseous or liquid HCl, the piping must also be designed with cold-resistant materials to ensure mechanical integrity under cryogenic conditions.

Risk #3

To mitigate the risk of fluid leakage and environmental release (loss of containment), strict preventive maintenance and inspection protocols must be enforced. Additional safeguards may include housing the process equipment within a ventilated enclosure equipped with scrubbers to capture fugitive emissions. Furthermore, installing sensors capable of continuously monitoring the concentrations of all major process components will enable rapid detection of abnormal levels. If elevated concentrations are observed, alarms can be activated to prompt immediate investigation and corrective action by plant personnel.

4. HAZOP Study

A Hazard and Operability (HAZOP) study was conducted on the T-100 distillation section to identify potential process deviations, their causes, and the consequences of these deviations during both normal and malfunctioning operating conditions. By dividing the system into logical ‘nodes’ and applying standard guidewords such as ‘less, ‘no,’ high,’ and ‘low’ to critical operating parameters, credible process hazards were evaluated in detail. Focus was placed on scenarios that may lead to loss of containment, unsafe operating conditions, environmental releases, equipment damage, and impacts on product quality. The goal of this analysis is to assess the adequacy of existing safeguards and recommend additional protections to ensure the safe, reliable, and compliant operation of the column and its associated equipment.

Table 11: *Node 1 - Feed to T-100 Column HAZOP*				
Deviation	Possible Causes	Consequences	Existing Safeguards	Additional Safeguards
No Flow	• Feed pump trip. • Upstream unit shutdown. • Control valve failed to shut.	• Column pressure and temperature fall. • Loss of separation. • Downstream product out of specification. • Potential column vacuum and air ingress, which could lead to flammability of vinyl chloride.	• Flow control with an alarm. • Upstream low-pressure trip.	• Install a low-flow interlock to place the column in safe standby. • Vacuum breaker. • Procedure for a controlled shutdown.
More Flow	• Flow controller failed at high. • Operator error in setpoint. • Pump running at the wrong speed. • Wrong pump in use.	• Flooding in the column. • Increased pressure. • Off-spec overhead and bottoms. • Potential relief valve lifting with venting of toxic VCM.	• Relief value on column. • High-level alarm in the reflux drum. • Flow controller.	• High feed flow trip. • Pressure trip on column. • Alarm for approach flooding.
Less Flow	• Partial blockage. • Pump degradation. • Partially closed valve. • Low upstream level.	• Column operating is unstable. • Poor separation. • Composition being off specified. • Potential polymer formation.	• Flow control with alarm.	• Predictive maintenance for pumps. • Quality monitoring.
Wrong Composition	• Upstream malfunction. • Incorrect manual. • Contamination. • Back-flow.	• Poor separation. • Accumulation of heavy ends or light non-condensable. • Increased corrosion. • Risk of explosion.	• Analyzers on feed and products. • Material balance monitoring.	• Interlock to divert off-spec feed to a safe holding tank

Table 12: *Node 2 - T-100 Column Bottom Section and Reboiler HAZOP*				
Deviation	Possible Causes	Consequences	Existing Safeguards	Additional Safeguards
High Temperature	• Reboiler stream valve failed to open. • The temperature controller is malfunctioning. • Loss of reflux due to fouling, causing poor heat transfer control.	• Overheating of a heavy VCM-containing mixture. • Thermal degradation and polymerization. • Overpressure. • Release via relief valve of toxic and flammable gases.	• Temperature control loop. • PSV on column. • Steam supply pressure control.	• Inhibitor injection to suppress polymerization. • Reboiler outlet temperature alarms.
Low Temperature	• Steam supply failure. • Steam valves closed. • Condensate back-up. • Excessive reflux.	• Loss of driving force for separation. • Poor stripping of light components. • Off-spec bottoms. • Possible accumulation of light VCM in the bottom system.	• Temperature controller. • Steam low-pressure alarm.	• Redundant steam supply or backup heating. • Automatic feed cutback on low bottom temperature. • Automotive product quality monitoring.
No Reboiler Flow	• Reboiler circulation pump trip. • Blocked reboiler tubes. • Isolation valve left closed after maintenance.	• Localized overheating in the reboiler. • Stagnant fluid and polymer buildup. • Sudden pressure rises. • Potential tube rupture.	• Level control in the column bottom. • PSV on reboiler and column. • Pump low-flow protection.	• Low-flow interlock to trip the steam • Hot work/startup checklist to verify valves open • Differential temperature/pressure monitoring across the reboiler

High Bottom Level	• Bottoms pump trip. • Outlet Valve is closed. • Over-refluxing. • Incorrect level setpoint.	• Liquid carrying over into the reboiler return or trays, causing flooding. • Reduced separation. • Possible liquid entrainment into the relief system during overpressure.	• Level controller with high-level alarm	• High-high level trip on feed or steam • Redundant bottoms pump or gravity drain to the safe tank • Overflow line routed to a contained sump/dike
Low Bottom Level	• Excessive bottom withdrawal • Faulty level indicator • Leak in bottom piping	• Reboiler starved of liquid, leading to tube overheating and failure • Potential fire/explosion if hydrocarbon leaks to hot surfaces	• Level controller. • Low-level alarm.	• Low-level trip. • Pump speed control. • Calibration of level instruments.

***Table 13**: Node 3 - T-100 Overhead Section, Condenser, and Reflux Drum HAZOP*

Deviation	Possible Causes	Consequences	Existing Safeguards	Additional Safeguards
High Pressure	• Condenser duty lost (cooling water failure) • Blocked vent; non-condensable build-up • The control valve to the reflux drum failed to close	• Relief valve lifting with release of toxic, flammable VCM • Potential rupture of column top or condenser • Fire/explosion risk	• PSV on column/overhead • Pressure control loop • Cooling water low-flow alarms	• High-high pressure trip to shut feed and reboiler • Independent overhead pressure alarm in the control room • Redundant condenser or backup cooling source
Low Pressure	• Excessive condenser duty	• Air ingress leading to oxygen in VCM	• Pressure control with minimum setpoint	• Nitrogen blanketing of the reflux drum

	• Vacuum from sudden cooling • Vent valve stuck open • Over-withdrawal of vapor	system leading explosive mixture • Poor column performance	• Vacuum breaker	• Low-pressure alarm and interlock to close vent • O_2 analyzer in overhead system
No Reflux	• Reflux pump trip • Reflux control valve closed • Low level in the reflux drum	• Loss of internal liquid traffic; sharp change in composition; overhead product extremely rich in light components; potential overpressure in column bottom	• Reflux flow controller • Low-level alarm in reflux drum	• High-high column temperature at upper tray trip to reduce feed/reboiler duty • Redundant reflux pump • Auto-start of standby pump
High Reflux Flow	• Reflux controller failed high • Operator error	• Column flooding; increased pressure; off-spec overhead and bottoms; potential for relief lifting	• Flow controller • Pressure control	• High-high differential-pressure across trays alarm • Reflux flow high-limit constraint in the control system
High Reflux Drum Level	• Product pump trip • Outlet valve closed • Incorrect setpoint • Condensed rain-in or water ingress	• Liquid carryover to vent/flare; potential emission of VCM and HCl; overflow to diked area	• Level controller and high-level alarm • Dike around the drum	• High-level trip to shut feed/reboiler and open diversion to emergency storage • Level switch hardwired to pump permissive
Low Reflux Drum Level	• Excessive withdrawal to products leak • Insufficient condenser duty	• Loss of reflux; pump cavitation; gas blow-by to pumps and downstream lines	• Low-level alarm • NPSH margin	• Low-level trip to stop reflux and product pumps Online leak detection • Periodic inspection of the drum and nozzles

Table 14: *Node 4 – Overhead Product from T-100 to Storage HAZOP*				
Deviation	Possible Causes	Consequences	Existing Safeguards	Additional Safeguards
No flow to storage	• Product pump trip • Downstream valve closed • Tank high-level interlock active • Line blockage	• Back-up of liquid in the reflux drum leading to high level, possible overflow, or venting to flare • Forced column rate reduction or shutdown	• Level control on the reflux drum • Tank high-level alarms • PSV/vent system	• Standby product pump with auto-start • Emergency diversion line to surge tank • Operator procedure for rapid controlled cutback
Reverse flow from storage	• Failure of the check valve • Pressure in the storage tank is higher than the column • Incorrect nitrogen padding	• Contamination of the column with storage tank contents • Overfilling of column/reflux drum • Potential oxygen ingress	• Check valves • Differential pressure control	• Periodic testing of check valves • Non-return valves in series • High-pressure alarm on storage tank • Interlock preventing over-pressuring the tank with nitrogen
Wrong destination	• Misaligned manual valves • Line connection error after maintenance	• VCM sent to the wrong tank or unit • Possible incompatibility (e.g., with water or other chemicals), causing polymerization,	• Line labeling; operating procedures • Lock-out/tag-out	• Positive line identification and color-coding • Use of key-interlocked valves • MOC and pre-startup safety

		corrosion, or overpressure		review (PSSR) for any tie-ins

Table 15: *Node 5 – Bottoms Product/ Heavy ends from T-100*				
Deviation	Possible Causes	Consequences	Existing Safeguards	Additional Safeguards
No flow from the bottom	• Bottoms pump trip • Valve closed • Downstream blockage	• Rising bottom level leading to column flooding • Entrainment • Possible overflow to the relief/vent system	• Level control • High-level alarm	• Redundant bottoms pump with auto-start • High-level trip to reduce feed/steam • Emergency gravity drain to the contained tank
More flow	• Pump speed too high • Controller failed high • Valve fully open	• Insufficient liquid level in the column bottom • Reboiler starvation • Tube overheating	• Level control • Low-level alarm	• Low-low level trip shutting steam and/or pump • Pump minimum-flow recycle line
Bottoms too light (off-spec composition)	• Reboiler duty too high • Incorrect reflux ratio • Tray damage	• Excess VCM and light components in the heavy-end system • Potential vaporization and pressure build-up in downstream equipment • Emission risks	• Periodic sampling • Temperature profiles along the column	• Online composition analyzer • Advanced process control to maintain spec • Alarm/interlock based on off-spec reading

The HAZOP review of the T-100 distillation system identified several high-risk deviations, including overpressure, overheating, polymer formation, and oxygen ingress, all of which could lead to the release of toxic and flammable vinyl chloride or damage to critical equipment. While existing safeguards, such as pressure relief valves, level control systems, and alarms, provide a foundation of protection, additional layers, including high-integrity interlocks, improved leak and composition monitoring, and enhanced shutdown procedures, are recommended to further reduce the likelihood and severity of hazardous events. Overall, the study reinforces that continued focus and operability, robust instrumentation, and proactive maintenance are necessary to ensure the long-term safety and reliability of the T-100 system.

5. AI INTEGRATION

5.1 AI Input and Output

Input:

I'll give you your prompt but carefully consider the project overview pasted "Vinyl-Chloride Process Safety and Environmental Hazard Analysis

Overview: Your consulting team has completed a preliminary analysis of the vinyl-chloride process. The

objective now is to conduct a thorough investigation focused on identifying and mitigating safety and

environmental risks associated with this process. The process flow diagram and stream table, provided as

Figure 1 and Table 1, outline the essential operating parameters and stream compositions. You objective

is to deliver a comprehensive safety and environmental hazard assessment report for the vinyl-chloride

process, including specific analysis and recommendations for the T-100 process units.

Output:

Vinyl-Chloride Process Safety and Environmental Hazard Analysis

AI Tool Used: ChatGPT (OpenAI), model **GPT-5.1 Thinking**
Date of use: 01 Dec 2025

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1. Introduction

This report presents an AI-assisted safety and environmental hazard analysis for a **vinyl-chloride production process**. The process flow diagram (PFD) and stream table (Figure 1 and Table 1 in the project handout) define the main equipment and operating conditions.

The objectives of this analysis are to:

- Identify **environmental hazards** (air, water, and solid wastes).
- Identify **material hazards** and **operational hazards** throughout the process.
- Summarize key **permissible exposure limits (PELs)** for critical toxic chemicals, especially vinyl chloride and hydrogen chloride.
- Perform a focused **HAZOP (Hazard and Operability) study** for the **T-100 process units**.

For this report, T-100 is treated as a **distillation column system for vinyl-chloride purification**, including feed, column, reboiler, condenser, accumulator, and product streams. Actual numerical conditions should be taken from the provided PFD and stream table and inserted where appropriate.

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2. Process Overview and Scope

2.1 General Process Description

The vinyl-chloride process typically involves:

1. **Feed preparation and reaction** – e.g., chlorination/oxy-chlorination of ethylene to produce ethylene dichloride (EDC), followed by cracking of EDC to vinyl chloride monomer (VCM) and HCl.
2. **Separation and purification** – distillation and phase-separation steps to isolate VCM from unreacted feeds, heavies, lights, and by-products.
3. **Recycle and utilities** – recovery of unreacted materials, off-gas handling, and integration with utilities (steam, cooling water, refrigeration).
4. **Storage and loading** – storage of VCM as a liquefied gas and loading to downstream units or transport.

2.2 T-100 Process Units

Within this flowsheet, **T-100** represents a **key separation unit** for VCM purification. Typical equipment around T-100:

- Feed drum and **feed pump**.

- **T-100 distillation column** operating near VCM boiling conditions (elevated pressure to maintain liquid phase).
- **Reboiler** providing heat input.
- **Overhead condenser** and **accumulator** managing condensed VCM and non-condensables.
- **Bottoms product** stream returning heavies or recycle.

Because T-100 handles **large inventories of volatile, toxic, and flammable VCM**, it is a critical focus for both **process safety** and **environmental protection**.

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3. Environmental Hazard Assessment

Environmental hazards are classified into **air emissions**, **water/wastewater contamination**, and **solid/hazardous wastes**.

3.1 Air Emissions

Primary pollutants:

- **Vinyl chloride monomer (VCM)** – volatile, flammable, and a known human carcinogen associated with liver angiosarcoma and other cancers. cancer.gov+2NCBI+2
- **Hydrogen chloride (HCl)** – corrosive, acid gas; irritant to respiratory system. NJ.gov+1
- Other volatile organic compounds (VOCs): unreacted EDC, light hydrocarbons, chlorinated by-products, and solvents (if present).

Potential emission sources:

1. **Fugitive emissions**
    - Pump seals, compressor seals, valve stems, flanges, sample connections, and loading arms handling VCM or chlorinated streams.
    - Particularly important around the **T-100 column, condenser, and accumulator**.
2. **Process vents and relief devices**
    - Overhead vent from T-100 condenser/accumulator.
    - Pressure relief valves (PRVs) from T-100, reboiler, and associated vessels, which may discharge to atmosphere if not routed to a flare or thermal oxidizer.
    - Vacuum system vents or purge gas streams.
3. **Combustion sources**
    - Boilers, heaters, or incinerators generating combustion products (NOx, CO, CO_2).
    - If VCM-containing streams are burned, potential formation of **dioxins and other toxic combustion products**, which are under increasing regulatory scrutiny. The Guardian+1

Consequences:

- Chronic health effects for workers and nearby communities due to long-term VCM exposure (cancer and liver damage). cancer.gov+2ATSDR+2
- Acute health effects from inhalation of HCl and high-concentration VCM during releases (respiratory irritation, CNS depression). NCBI+1
- Ambient air quality violations and regulatory non-compliance.

Recommended controls:

- **Closed-vent systems** from T-100 overheads and PRVs routed to a flare or thermal oxidizer.
- **Refrigerated or sub-cooled VCM systems** to minimize vapor pressure and emissions.
- **LDAR (Leak Detection and Repair) program** using portable VOC detectors and regular inspections of valves, pumps, and flanges.
- Use of **double mechanical seals** with buffer systems on pumps handling VCM.
- Real-time **fixed gas detectors** and alarms for VCM and HCl in critical areas (around T-100 and storage).

3.2 Water and Wastewater Hazards

Potential contamination pathways:

1. **Cooling water system leaks**
   - Tube failure in T-100 condenser or other heat exchangers can contaminate cooling water with VCM and chlorinated organics.
   - Contaminated cooling water discharged to surface water or a cooling tower can cause air stripping of VCM and environmental release.
2. **Process drains and stormwater**
   - Floor drains in process areas around T-100 may collect liquid spills or routine washdown, leading to contaminated wastewater.
   - Rainwater run-off from VCM handling areas.
3. **Wastewater from equipment cleaning and maintenance**
   - Solvent or water flushes used to decontaminate T-100 and associated equipment during shutdowns.

Consequences:

- Toxicity to aquatic life and potential bioaccumulation of chlorinated organics.
- Regulatory violations for discharge permits (e.g., NPDES) if limits for VCM and related compounds are exceeded.

Recommended controls:

- **Segregated, contained drainage** in VCM and T-100 areas routed to a dedicated **process wastewater collection** and treatment system.

- Secondary containment and curbing to keep spills out of stormwater drains.
- **Heat exchanger integrity management** (periodic inspection, pressure testing, leak detection) and rapid isolation procedures for suspected leaks.
- On-site treatment steps such as **stripping or activated carbon** to remove VOCs from wastewater before discharge.

3.3 Solid and Hazardous Wastes

Sources:

- **Off-spec VCM product** and contaminated residues from T-100 column bottoms or reboiler.
- **Spent adsorbents, filters, and catalyst residues** from upstream units.
- Sludge from wastewater treatment containing chlorinated organics and heavy components.

Hazards and controls:

- Many of these wastes are likely **ignitable or toxic**, qualifying as hazardous wastes under RCRA; they must be properly characterized and disposed of via permitted incineration or secure landfill.
- Designated **waste accumulation areas** with secondary containment, labeling, and documentation.
- Preference for **recycling or reprocessing** off-spec VCM and intermediates where technically feasible to minimize hazardous waste generation.

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4. Material and Operational Hazard Assessment

4.1 Key Hazardous Materials and PEL Summary

Table 1 summarizes key materials in a typical vinyl-chloride process and selected exposure limits. Exact materials and compositions should be checked against your stream table and SDSs.

Table 1 – Example hazardous materials and exposure limits

Substance	Main hazards (brief)	OSHA PEL (US)	Type	Notes
Vinyl chloride	Flammable gas; CNS depressant; **human carcinogen** (liver angiosarcoma, etc.). cancer.gov+2NCBI+2	**1 ppm** 8-hr TWA; **5 ppm** not to be exceeded in any 15-min	TWA, STEL	OSHA standard 1910.1017; carcinogen – exposures should be kept as low as feasible.

		period OSHA+1		
Hydrogen chloride (HCl)	Corrosive gas; severe respiratory irritant; forms hydrochloric acid on contact with moisture. NJ.gov+1	**5 ppm** ceiling (C) – not to be exceeded at any time OSHA+1	Ceiling	Often present as HCl gas or dissolved acid.
Chlorine (if present)	Strong oxidizer; toxic inhalation hazard; corrosive to eyes and lungs.	0.5 ppm TWA; 1 ppm STEL (annotated PEL table) OSHA	TWA/STEL	Depending on process route.
EDC and other chlorinated organics	Flammable/toxic VOCs; potential liver and CNS effects.	Varies by compound (consult SDS and OSHA Table Z-1). OSHA	–	Treat as hazardous VOCs requiring strict containment.

Note: Values above are representative and based on current OSHA and state/NIOSH references as of 2024; actual design and plant operation must refer to the latest regulations and safety data sheets.

4.2 Operational Hazards by Process Area

1. **Reaction Section**
   - Exothermic reactions, risk of **runaway** if cooling fails.
   - Overpressure due to rapid gas generation.
   - Handling of highly reactive and toxic feeds (e.g., Cl_2, HCl).
2. **T-100 Separation System (focus area)**
   - Large inventory of **VCM** at elevated pressure and near its boiling point.
   - Heat input in reboiler and condensation at the top create potential for:
     - Overpressure if vent paths blocked.
     - High temperature excursions if level control fails or reboiler overheats.
   - Composition control failures may lead to high impurities in product or accumulation of heavies/light ends.
3. **Storage and Loading**
   - Overfilling of VCM storage tanks or transport vessels.
   - Hose or loading arm failures during truck/railcar loading.
   - Cold-temperature embrittlement or excessive pressure due to warming of stored liquid VCM.
4. **Utilities and Common Systems**
   - Loss of power or instrument air leading to loss of control and interlocks.
   - Cooling water or refrigeration failures increasing temperature and pressure in VCM equipment.

Typical safeguards:

- Process control systems (temperature, pressure, level controllers).
- **Pressure relief valves (PRVs)** routed to flare.
- High-high trip interlocks for key variables (e.g., column pressure and level).
- Gas detection and emergency shutdown procedures.
- Operator training and written operating procedures for startup, normal operation, and shutdown.

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5. HAZOP Study for T-100 Process Units

The following HAZOP is a qualitative example focusing on the most significant deviations for T-100. Students should refine this with actual conditions from the project PFD and stream table.

5.1 Nodes Considered

1. **Node 1 – Feed to T-100** (feed drum and pump discharge to column).
2. **Node 2 – T-100 Column (mid-section)**.
3. **Node 3 – Reboiler loop**.
4. **Node 4 – Overhead condenser and accumulator**.
5. **Node 5 – Bottoms product line.**

5.2 Guidewords and Variables

- **Guidewords:** No, More, Less, Reverse, As well as, Other than.
- **Variables:** Flow, Temperature, Pressure, Level, Composition.

5.3 Example HAZOP Tables

Node 1 – Feed to T-100 (feed line and pump)

Deviation	Possible Causes	Consequences	Existing Safeguards	Recommendations
No flow	Feed pump trip; pump suction blocked; valve closed; control valve failure closed.	Column inventory and separation collapse; loss of reflux balance; potential off-spec product; reboiler overheating.	Flow controller with low-flow alarm; pump low-suction trip.	Add automatic column heat-input reduction on low feed; procedure for rapid safe shutdown.
More flow	Control valve failure open; incorrect	Column flooding; loss of separation; high pressure	Flow controller; high-level	Add high-high column differential-pressure alarm; consider feed flow high-high trip.

	setpoint; manual bypass open.	drop; potential PRV lift.	alarm in column.	
Wrong composition	Upstream process upset; wrong stream routed; contamination from utilities.	Column design assumptions invalid; unexpected heavies or lights accumulating; potential polymerization or fouling.	Feed analyzers (if present); operator checks.	Periodic online composition checks; interlock on out-of-spec composition for critical contaminants.

Node 2 – T-100 Column (mid-section)

Deviation	Possible Causes	Consequences	Existing Safeguards	Recommendations
High pressure	Condenser failure; vent blocked; excessive feed or vapor rate; PRV stuck closed.	PRV lift to flare or atmosphere; potential rupture; large VCM release.	PRV to flare; high-pressure alarm.	High-high pressure trip closing feed and cutting heat; ensure relief routed to safe system.
Low pressure	Excessive condensation; vacuum created during shutdown; vent valve open.	Air ingress and formation of explosive mixture; mechanical damage (buckling).	Vacuum breaker; operating procedures.	Confirm inert gas purge during shutdown; add low-pressure alarm.
High temperature (tray)	Too much reboiler duty; loss of reflux; low feed rate.	VCM overheating; possible accelerated decomposition or unwanted reactions; product off-spec.	Temperature control on reboiler; alarms.	Add high-high tray temperature trip or alarm; procedures for response.

Node 3 – Reboiler Loop

Deviation	Possible Causes	Consequences	Existing Safeguards	Recommendations
No circulation	Reboiler pump failure; valve closed; fouling blockage.	Hot spots, film boiling; potential tube failure; column loses separation and level control.	Pump protection; low-flow alarm.	Provide standby pump; regular cleaning schedule; differential-pressure monitoring.

More heat	Steam valve failure open; control malfunction.	Excess vaporization; high column pressure; overhead system overload.	Temperature controller; PRV.	High-high column pressure trip that closes steam; periodic testing of control valves.

Node 4 – Overhead Condenser and Accumulator

Deviation	Possible Causes	Consequences	Existing Safeguards	Recommendations
No condensation	Loss of cooling water or refrigerant; condenser fouling.	Overhead vapor cannot condense → rising column pressure and temperature; PRV may lift, releasing VCM to flare or atmosphere.	Cooling water low-flow alarm; PRVs.	Add automatic heat-input cutback on high column pressure; backup cooling or emergency procedures.
High accumulator level	Reflux control valve failure closed; high product withdrawal.	Liquid carry-over to downstream system; risk of sending slugs of VCM to flare or compression system; flooding of column top trays.	Level controller with alarm.	High-high level trip closing feed or product; ensure adequate knockout capacity upstream of flare.
Low accumulator level	Excessive reflux or product withdrawal; leak.	Loss of reflux → reduced separation, high overhead temperature, off-spec VCM; potential overheating of column.	Level controller.	Low-low level alarm and interlock to reduce product draw; investigate leak detection.

Node 5 – Bottoms Product Line

Deviation	Possible Causes	Consequences	Existing Safeguards	Recommendations
No flow	Bottoms pump trip; valve closed; line blocked.	Rising column bottom level; risk of flooding; liquid carry-over into reboiler and other equipment.	Level control and alarm on column.	High-high bottom level trip; provide spare pump.
Reverse flow	Back-pressure from downstream unit or storage.	Contamination of T-100 with downstream material; potential unexpected reaction.	Check valves; pressure control.	Periodic check of non-return valves; install additional isolation as needed.

6. Overall Risk Mitigation and Recommendations

Based on the environmental, material, and HAZOP analyses, the most significant risks for the vinyl-chloride process—especially around T-100—are:

- **Chronic and acute exposure to VCM**, a known human carcinogen, through fugitive emissions, vent releases, or catastrophic failures. cancer.gov+2NCBI+2
- **Corrosive and toxic exposures to HCl** and other acid gases. NJ.gov+1
- **Overpressure and loss of containment** events in T-100 and its associated equipment, leading to large atmospheric releases and potential ignition.
- **Environmental contamination** of water and soil through uncontrolled wastewater and hazardous waste handling.

High-priority recommendations:

1. **Ensure compliance with OSHA PELs and process safety regulations** for vinyl chloride and other hazardous chemicals, including implementation of a comprehensive **PSM (Process Safety Management)** program where threshold quantities are exceeded. OSHA+1
2. **Route all VCM and HCl relief and vent streams** from T-100 and related equipment to a **flare or thermal oxidizer** with appropriate knockout and treatment systems.
3. Implement robust **LDAR** and **fixed gas detection** around the T-100 area and storage facilities.
4. Strengthen **instrumented protection layers** (interlocks and trips) for:
   - High column pressure,
   - High and low levels in critical vessels,
   - Abnormal temperatures in the column and reboiler.
5. Provide **segregated drainage, spill containment, and wastewater treatment** for all VCM-handling areas.
6. Develop and drill **emergency response plans** for toxic gas releases, fires, and explosions, including community notification and shelter-in-place procedures.

Medium-priority recommendations:

- Optimize column control strategy (feed, reflux, and reboiler duty) for stability during upsets.
- Increase **online monitoring of compositions** for critical streams (e.g., VCM purity, heavies/light ends).
- Improve documentation and training: accessible standard operating procedures, HAZOP findings integrated into operator training, and periodic refresher courses.

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7. Conclusions

Using ChatGPT (GPT-5.1 Thinking) as the AI tool, a qualitative yet comprehensive evaluation of the **vinyl-chloride process safety and environmental hazards** has been developed with particular emphasis on **T-100 process units**.

The analysis covers:

- Major **environmental hazards** (air emissions, wastewater, and solid waste) and practical mitigation strategies.
- Key **material hazards** and relevant **OSHA exposure limits**, emphasizing the carcinogenic nature of vinyl chloride and corrosivity of hydrogen chloride.
- Principal **operational hazards** across the process.
- A focused **HAZOP study** for T-100, highlighting credible deviations, causes, consequences, and safeguards.

This AI-generated report should be reviewed and supplemented by the student team with **process-specific data from the PFD and stream table, local regulatory requirements, and instructor feedback**. It is not a substitute for professional engineering judgment, but it provides a structured starting point and a consistent framework for the final project report.

5.2 AI Evaluation

For this report, it was imperative for the team to explore the usage of AI in process safety as the technology has made headway especially in engineering applications and determined it'd be appropriate in considering its input in our general decision-making process. The AI platform used was ChatGPT by OpenAI using its version 5.1 Thinking GPT.

The AI was decent at getting the report structure and especially seen in rightly going over the various categories ranging from the environmental, material, and operational hazards, PELs and a HAZOP. The details were laid out in a logical order that looked appropriate for a safety report.

The biggest pro to the AI's use was its organization as it helped separate the air, water and solid-waste issues and tied them back to the T-100 and listed the safeguards like flares, relief valves, LDAR programs, interlocks and gas protectors then connected them to the hazards they were required to protect. Also, another positive was the HAZOP tables. The causes, consequences and deviations showed things that were mentioned to look for in class and the AI put it into a table that was easy to read.

The AI also assisted with the safety regulatory side by noting that vinyl chloride is a carcinogen and that OSHA has a low PEL for it, and the process would most likely be categorized under OSHA PSM. It was believed that this might not have been fully considered but the AI stressed the need for more multiple protection layers making the report sound more cohesive and constantly brought up emergency response planning and wastewater separation which can be lost when the focus is on equipment.

Despite the positives, there were obvious limits. The AI didn't have access to specific PFD and stream tables, so the numbers were assumed and are generic. Assumptions include T-100 is simply a VCM purification column and guessed for the typic hazards but couldn't accurately confirm temperatures, pressures, or compositions for the actual project, so the recommendations made aren't tailored to the conditions in the assignment. To make sure there weren't contradictions, the project handouts had to be reviewed.

Another issue was the AI sounded confident when it was clearly guessing and would say things unless you keep questioning and could write lengthy paragraphs with buzzwords.

Overall, the AI was useful as a tool for brainstorming with help on the outline, decent hazard categories and entries for sample HAZOP which helped with time and keeping focus on some important topics. However, it is still a long way to go as it didn't understand processes and needs human engineering to verify regulatory limits and edit the HAZOP to fit operating conditions. AI would be great for early drafts, but not as a final say on decisions for process safety.

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